HaoChen Xia

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Education

University of Illinois at Urbana Champaign **Graduation Date 2026**

Major in Computer Engineering with Minor in Mathematics GPA: 4.0/4.0

- Dean's List: Fall 2022, Spring 2023, Fall 2023, Spring 2024
- Relevant coursework: Computer System Engineering, Machine Learning, Algorithm, Artificial Intelligence, Data Structure, Robotics, Probability with Engineering Application, Linear Algebra, Data Science, Fundamental Mathematics

Publications

1. S Liu, **H Xia**, FC Pouria, K Hong, N Chakraborty, K Driggs-Campbell, "HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments", under review, [Paper, Website, Video]

Research Experience

Human-Centered Autonomy Lab Champaign, IL

Advisor: Professor Katie Driggs-Campbell

Undergraduate Research Assistant February 2024 – Present

- **HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments**
  - Collaborated to develop and evaluate Heterogeneous Interaction Graph Transformer (HEIGHT)
  - Applied A* Search Algorithm to HEIGHT to analyze model architecture in ablation study
  - Developed neural networks baseline without attention mechanisms to investigate the effects of spatio-temporal interaction graph used in deep reinforcement learning policies
  - Implemented dynamic window approach on a low-fidelity simulator as a non-neural-network baseline
- **Ensemble-Based Classification Framework for Improved Gesture Recognition**
  - Developed and implemented an ensemble-based classification framework to enhance the model performance
  - Assisted in exploring various unsupervised learning and sampling methods to optimize class subset formation
  - Conducted experiments and ablation studies on three datasets to evaluate the framework and its components
- **Generative Model for In-car Hand Gesture Video Generation**
  - Collaborated to develop variational autoencoder (VAE) and generative adversarial network (GAN) to generate in-car hand gesture videos
  - Investigated neural network architecture and parameters to optimize reconstruction video generation models
  - Developed transformer-based classifiers to classify frames in the latent space to evaluate the quality of latent space

Research in ECE Department at UIUC Champaign, IL

Advisor: Professor Deming Chen and Professor David M. Nicol

Undergraduate Research Assistant January 2024 – Present

- **High-Performance Fault Tolerant Communication Protocols in Safety-Critical Industry**
  - Developed and implemented Data Plane Development Kit (DPDK) compatible versions of Parallel Redundancy Protocol (PRP) to investigate the optimization of PRP
  - Simulated complex network environments using Omnet++ and C++ to evaluate the effectiveness of applying PRP to critical nodes in the network
  - Collaborated to construct a redundancy network to evaluate the performance of conventional and optimized PRP
  - Analyzed the performance bottlenecks of the conventional PRP and improvements achieved by DPDK

Working Experience

Nokia Naperville, IL

5G Arch January 2025 – Present

- Collaborated to improve the OpenSearch index with nested fields to enhance multi-vector support for k-NN search
- Enabled image attachments and image-based search functionality in the chatbot and search engine
- Applied Vision Language Model in data transformation between DOORS and OpenSearch for image processing

Garg Group at UIUC Champaign, IL

Computer Vision Engineer May 2024 – Aug 2024

- Developed real-time object tracking and counting algorithms using Deep Simple Online and Real-time tracking (DeepSORT) to accurately count specific objects in video footage
- Applied decision-level multi-modal techniques to improve the performance of You Only Look Once (YOLO) models and overall algorithms
- Achieved data auto-annotation by combining YOLO with Segment Anything Model (SAM)
- Collected and annotated waste data using Roboflow to train YOLO models for detecting different types of waste

Applied Technologies for Learning in the Art & Science Internship Program Urbana, IL

Data Analyst & Educational Technologist January 2023 – May 2023

- Managed and visualized K-12 educational program data from 70+ schools to evaluate the effectiveness of the program
- Analyzed data from 100+ survey questions before and after the program to assist in developing educational strategies
- Created videos and Word tutorials to teach clients to analyze and visualize data with Power BI and Tableau

Leadership Experience

1Shan AI Team Hybrid, IL

Cofounder & Tech Lead March 2024 – Present

- **Mythoverse AI – AI-Powered Video Storytelling**
  - Developed an AI-powered text-to-video generation system with advanced semantic understanding for high-quality video creations with various LLM and VLM, such as GPT and FLUX
  - Collaborated to implement real-time AI-generated voiceovers, subtitles, and sound effects to enhance video storytelling with TTS AI
  - Designed and developed an interactive platform for users to engage with Mythoverse AI using Django
- **1Shan AI – Aesthetic Intelligence for Visual Content**
  - Collaborated to develop AI-driven an aesthetic model to evaluate, rank, and enhance clothing visuals to enable personalized and high-quality custom apparel design
  - Assisted in implementing reinforcement learning fine-tuning to refine text-to-image generation outputs to ensure consistently high-quality visuals
  - Designed and developed an interactive website for users to engage with 1Shan AI using Django
- **Automated Codebase Modification with Large Language Model**
  - Analyzed the stages and workflows of the Warehouse Management System codebase written in Django and Vue.js
  - Investigated Repo-agent and SWE-agent to enable automated modification of the codebase by a large language model
  - Developed and optimized suitable prompt inputs and configurations for large language models to ensure accurate and effective codebase modification

Project Highlights

Keylogger with Large Language Model Urbana, IL

Independent project April 2023 – July 2024

- Developed a keylogger in Rust to record computer activities, including files, actions, and keyboard inputs
- Designed and implemented filtering algorithms to identify and highlight potentially sensitive information
- Implemented encryption and decryption algorithms to encode and decode recorded data for disguise purpose
- Enhanced keylogger result analysis by integrating large language models

UR3 Robot Arms Tools Recognition and Handling in Collaborative Work Environments Urbana, IL

ECE470 Final Project March 2024 – May 2024

- Developed a real-time algorithm for UR3 Robot Arms to classify tools, determine optimal pick-up points, and accurately deliver tools to human hands
- Implemented and integrated computer vision techniques to analyze tools' shapes and positions to enhance robot arms' ability to assist human work
- Evaluated and optimized the algorithm's performance in dynamic environments to ensure continuous and accurate tool handling and delivery

SigAIDA Machine Learning and Computer Vision Projects Urbana IL
SigAIDA Project Team Lead August 2023 – February 2024

- Led a project team at SIGAIDA to build Region-Based Convolution Neural Networks (R-CNN) to detect dangerous objects in images
- Developed multiple Long Short-Term Memory (LSTM) models to predict stock prices on daily, weekly, and monthly intervals
- Collaborated on developing CNN models for classifying potato diseases, animals, and guns
- Implemented plant leaf classification using pretrained VGG19 and Imagenet50 models

A* Search Algorithm Implementation Urbana IL
CS225 Extra Credit Project October 2023 – December 2023

- Implemented A* Algorithm to find the shortest path in a graph from source to destination without traversing all the nodes
- Designed several heuristic functions to handle different graphs and situations
- Generated random grids and graphs to test the algorithm to ensure its correctness and time efficiency

Android Map App with Comments and Planning System Urbana, IL
CS124 Final Project October 2022 – December 2022

- Developed a comment system to give comments and rates to a place on an Android map of Urbana-Champaign with Java
- Created a planning system to add plans and notes to a place with a map pin on the map
- Implemented search algorithms to search for places on the map based on names and types

Honors

Yunni and Maxine Pao Memorial Scholarship 2024 – 2025
ECE Simmons Scholarship 2024 - 2025
Bradley A. Simmons Memorial Scholarship 2024 – 2025
Kaskowitz Family Scholarship 2024 – 2025
IL Engineering Outstanding Scholarship 2024 - 2025
James Scholar Honors Program 2023 - 2026
American Mathematics Competition 12 December 2021

- Certificate of Distinction

National Olympiad in Informatics in Provinces 2021 December 2021

- First Place

United States of America Computing Olympiad February 2021

- Silver Place

United States Academic Decathlon China 2021 February 2021

- National Individual Overall Award, Third Place
- Second Place in Math and Art
- National Overall Team Award, Third Place

Certificates

IBM AI Engineering Professional Certificate December 2023
IBM Building Deep Learning Models with TensorFlow Certificate December 2023
IBM Deep Neural Networks with PyTorch Certificate November 2023
IBM Introduction to Computer Vision and Image Processing Certificate November 2023

Career Certificate – International Student October 2023

IBM Introduction to Deep Learning & Neural Networks with Keras Certificate October 2023

IBM Machine Learning with Python Certificate September 2023

Technical Skills

Programming language: C++/C, Python, Rust, Java, HTML, CSS, JavaScript, C

Packages: Pytorch, TensorFlow, OpenCV, Keras, Scikit-learn, OpenAI, Ultralytics, Django, DPDK, Omnet++, Inet

Software & Tools: LaTeX, ROS, RoboFlow, Git, OpenSearch, DOORS, IBM CV Studio, Power BI, Haskell, Tableau

Extracurricular Activities

Association for Computing Machinery Urbana, IL

Participant September 2022 – Present

- Volunteered and participated in Reflection | Projection activity to assist participants and listen to tech talks
- Expand computer knowledge about hacking and cybersecurity using Python in Special Interest Group Pwny

Institute of Electrical and Electronics Engineers Urbana, IL

Participant September 2022 – Present

- Participate in TAG, Flask, and PCB workshops to learn networking, information security, Flask, and PCB
- Meet with leaders from the industry in tech talks to learn applications and technology about electrical engineering

Leadership I Program Urbana, IL

Participant October 2022 - December 2022

- Developed leadership skills in innovation and problem-solving with other participants and facilitators
- Improved intrapersonal and interpersonal skills by completing self-knowledge and innovation inventory